

Olive oil and walnut breakfasts reduce the postprandial inflammatory response in mononuclear cells compared with a butter breakfast in healthy men.

BACKGROUND: Inflammation is crucial in all stages of atherosclerosis, and few studies have investigated the effect of dietary fat on markers of inflammation related to this disease during the postprandial period. **OBJECTIVE:** To evaluate the chronic effects of dietary fat on the postprandial expression of proinflammatory genes in peripheral blood mononuclear cells (PBMCs) in healthy subjects. **DESIGN:** 20 healthy men followed three different diets for 4 weeks each, according to a randomized crossover design: Western diet: 15% protein, 47% carbohydrates (CHO), 38% fat (22% saturated fatty acid (SFA)); Mediterranean diet: 15% protein, 47% CHO, 38% fat (24% monounsaturated fatty acid (MUFA)); CHO-rich and n-3 diet: 15% protein, 55% CHO, <30% fat (8% polyunsaturated fatty acid (PUFA)). After 12-h fast, volunteers were given a breakfast with a fat composition similar to that consumed in each of the diets-butter breakfast: 35% SFA; olive oil breakfast: 36% MUFA; walnut breakfast: 16% PUFA, 4% alpha-linolenic acid (LNA). **RESULTS:** The butter breakfast induced a higher increase in tumor necrosis factor (TNF)-alpha messenger RNA (mRNA) expression than the olive oil or walnut breakfasts ($P=0.014$) in PBMCs. Moreover, we found a higher postprandial response in the mRNA of interleukin (IL)-6 with the intake of butter and olive oil breakfasts than with the walnut breakfast ($P=0.025$) in these cells. However, the effects of the three fatty breakfasts on the plasma concentrations of these proinflammatory parameters showed no significant differences ($P=N.S.$). **CONCLUSION:** Consumption of a butter-enriched meal elicits greater postprandial expression of proinflammatory cytokine mRNA in PBMCs, compared to the olive oil and walnut breakfasts.